

Blood supply of the lung

Essay

1. How many bronchial arteries are there?

On the left side: Two arteries

On the right side: Single artery

2. Where do the bronchial arteries arise and what do they supply?

On the left side: from descending aorta

On the right side: from superior left bronchial artery

They supply bronchi, stroma of the lung, and the visceral pleura

3. Enumerate the pulmonary veins and what does it carry?

Superior pulmonary vein.

Inferior pulmonary vein.

They are two in each side carrying oxygenated blood from the lung.

4. Which arteries carry deoxygenated blood to the lung?

Right pulmonary artery: lies anterior and in between ep-arterial and hyp-artetrial bronchi.

Left pulmonary artery: lies antero-superior to the left main bronchus.

MCQ

1. Blood pressure in the pulmonary artery is _____.
a) Same as that in the aorta
b) Less than that in venae cavae
c) More than that in the carotid
d) More than that in the pulmonary vein.

ANSWER: (d)

2. Pulmonary veins differ from other veins in that they:
a) Carry blood away from the heart
b) Carry deoxygenated blood
c) Carry oxygenated blood
d) Have cartilaginous walls

ANSWER: (c)

3. What blood vessel carries deoxygenated blood from the heart to the lungs?
a) Pulmonary vein
b) Aorta
c) Pulmonary artery
d) Vena cava

ANSWER: (c)

MCQ

1-Which of the following represents the surface anatomy of the apex of the heart?

a-Lies opposite the right 5th intercostal space 9 cm from the middle line

b-Lies opposite the left 5th intercostal space 9 cm from the middle line

-c-Lies opposite the left 6th intercostal space

d-Lies opposite to 4th thoracic vertebrae

2-. Which of the following represents the parasympathetic innervation of the heart?

A. The vagus nerve

B. The intercostal nerve

C. The sympathetic chain

D. The facial nerve

3-the covering of the heart is:

E. a-the pleura

F. b-the pericardium

G. c-synovial membrane

H. d-mediastinum

4-the 4 pulmonary veins open into:

I. a-right atrium

J. b-right ventricle

K. c-left atrium

L. d-left ventricle

5-The left subclavian artery arises from

- a-Innominate artery
- b- the arch of the aorta**
- c- ascending aorta
- d- descending aorta

6-The right subclavian artery arises from

- a- the innominate artery**
- b- the arch of the aorta
- c- ascending aorta
- d- descending aorta

7-The left coronary artery:

- a-Arise from the anterior surface of the aorta
- b-Supplies the posterior interventricular branch
- c-Supplies the diaphragmatic part of right ventricle
- d-Pass between the pulmonary trunk and left auricle**

8-the pulmonary trunk arise from:

a-right atrium

b-right ventricle

c-left atrium

d-left ventricle

9-the apex of the heart formed by :

a-left ventricle

b-left atrium

c-right atrium

d-right ventricle

10-the blood supply of heart is:

a-arch of aorta

b-coronary arteries

c-bronchial arteries

d-pulmonary artery

short essay Q:

1- Enumerate the valves of the heart and its position?

2- describe the surface anatomy of the heart?

3-describe the external features of right atrium?

4-describe the external features of left atrium?

5-Enumerate the openings of right atrium?

6-describe the arterial supply of the heart?

7-enumerate the openings of left ventricle?

8-describe the external features of the left ventricle?

9- describe the external features of the right ventricle?

10-enumerate the covering of the heart?

Physio. Q- Level 4-CARDIOTHORACIC MEDICIN and SURGERY

DR. Dr. Shaymaa fouad

1. List four factors affecting pulmonary blood pressure and flow:

1. Automatic Control {Alveolar Oxygen}:
2. Cardiac output.
- 3 .Left atrial pressure
4. Gravity
- 5 .Nervous control
- 6 . Chemical substances
7. Respiratory movements

2. What is the meaning of Ventilation/Perfusion Ratio:

It denotes the ratio of alveolar ventilation and the amount of blood that perfuse the alveoli.
Mathematically, $V/Q = 0.84$

3- Why the pulmonary artery BP is lower than the systemic arterial BP ?

- a) The output of the right ventricle is less than that of the left ventricle.
- b) The left ventricular wall is thicker than that of the right ventricle. d
- c) The aorta is more elastic than the pulmonary artery.
- d) The pulmonary resistance is less than the systemic resistance.

4- Which of the following is true regarding pulmonary circulation?

- A. It is a low-pressure system that allows for improved gas exchange.
 - B. It is the larger of the two circulatory systems.
 - C. The system functions with an increased arterial pressure to circulate through the distal parts of the body.
 - D. It consists of the left side of the heart, the aorta, and its branches.
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1- Determine the ionic basis of ventricular action potential

Phase 0: Initial rapid depolarization: due to Na^+ inflow due to opening of fast Na^+ channels.

Phase 1: Brief initial repolarization: due to closure of Na^+ channels, little K^+ efflux and Cl^- influx.

Phase 2: Prolonged plateau: due to opening of slow prolonged Ca^{++} channels.(A balance is created between influx of Ca^{++} and outflux of K^+). So, the potential is kept at 0mV

Phase 3: Late rapid repolarization: due to prolonged opening of K^+ channels and closure of Ca^{++} channels.

Phase 4: Resting membrane potential is restored (-90 mv) by $Na^+ - K^+$ pump.

2- Demonstrate Relationship Between Heart Rate (HR), Cardiac Filling, and Cardiac Output (CO) Figure 6:

Excessive tachycardia ≥ 200 beats /min. decreases CO by $\downarrow$ preload and stroke volume.

It also impairs cardiac function from decreased coronary perfusion in shortened diastole.

Sever bradycardia ≤ 50 beats/minute decreases cardiac output. Although stroke volume $\uparrow$ but this increase will be insufficient to balance the excessive decrease in the heart rate.

Mild decrease from 70-50 beats/minute, the ventricles have enough time for maximum filling. So, the stroke volume $\uparrow$ and the cardiac output remains constant.

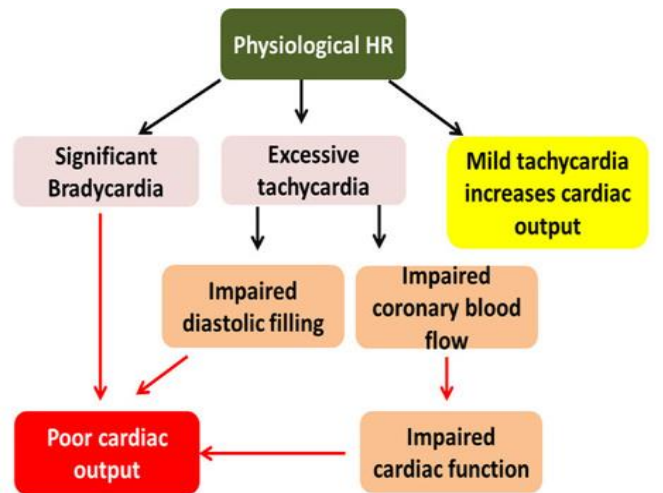


Fig.6
Relationship between heart rate (HR), cardiac filling and cardiac output (CO)

3- A 75-year-old woman with a history of chronic congestive heart failure undergoes cardiac catheterization to determine her extent of cardiac dysfunction. During estimation of her systolic function, in what phase of the cardiac cycle should her left ventricular volume unchanged?

- A. Rapid filling
- B. Isovolumetric contraction**
- C. Ventricular ejection
- D. Atrial systole

4- What is the cardiac ejection fraction?

- A. It is the percentage ratio of cardiac work to total energy expenditure.
- B. It is the cardiac output per square meter body surface area.
- C. It is the percentage of blood leaving the heart in each cardiac cycle.**
- D. It is a premature contraction of the heart occurring during diastole.

Physio. -CARDIOTHORACIC MEDICIN and SURGERY

DR. Azza Hamed AbdElwahab

1- About the pulmonary vascular resistance, Which of the following is incorrect?

- a. It is decreased during exercise.
- b. May lead to pulmonary hypertension.
- c. It is increased in lung diseases.
- d. **It is decreased in hypoxia.**

2- If the cardiac output is doubled, what happens to pulmonary arterial pressure?

- a. Becomes 2 times normal.
- b. **Kept constant.**
- c. Becomes 4 times normal.
- d. Markedly decreased.

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5- A 75-year-old woman with a history of chronic congestive heart failure undergoes cardiac catheterization to determine her extent of cardiac dysfunction. During estimation of her systolic function, in what phase of the cardiac cycle should her left ventricular volume unchanged?

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6- Why the S A node is the normal pace maker of the heart?

- (a) Is the most rapidly discharging part.
- (b) Is the most richly supplied by nerve endings.
- (c) Located in the atrium.
- (d) Is the most rapidly conducting part.

7- Which of the followings characterizes the S-A node?

- (a) Cells within the S-A node act as heart pace maker because their membrane depolarized to threshold and initiate an action potential.
- (b) Acetyl choline increases the slope of the pace-maker potential.
- (c) Sympathetic stimulation decreases the slope of the pace maker potential.
- (d) The pace-maker cells within the S A node are neurons rather than myocytes.

8-What is the cardiac tissue having slowest rhythmicity ?

- (a) SAN without vagal tone.
- (b) SAN with vagal tone .
- (c) Ventricular muscle.
- (d) AVN.

9- In which phase of cardiac cycle, the second heart sound can be heard?

- a) Atrial contraction phase.
- b) Isometric relaxation phase.
- c) Maximum ejection phase.
- d) Reduced filling phase.

10- Which of the following events occur during maximum ejection phase of cardiac cycle?

- a) The first heart sound is heard.
- b) The T-wave of the ECG is at its peak.
- c) The ventricular volume is constant.
- d) The ventricular pressure reaches maximum.

- 1. Enumerate factors affecting pulmonary blood pressure and flow**
- 2. What is the meaning of Ventilation/Perfusion Ratio:**
- 3. Determine the ionic basis of ventricular action potential**
- 4- What is the cardiac ejection fraction?**
- 5. Describe relationship between heart rate (HR), cardiac filling, and cardiac output (CO)**
- 6. What do you know about lung zones?**

Q-Type of phosphodiesterase enzyme inhibitors used in respiratory diseases with example?

DE-4 inhibitors: roflumilast

-PDE-5 inhibitors: sildenafil

-Non-specific PDE inhibitors: theophylline

Q- Adverse effects of sildenafil?

short duration and dose related

1. Headach
2. Flushing
3. Nasal congestion
4. Dyspepsia
5. Large dose inhibit PDE6 needed for phototransduction which cause blue vision, blurred vision , visual field contraction

Q- Adverse effects of Roflumilast?

1. Gut... nausea,vomiting, diarrhea, anorexia,pain
2. Adipose tissue .. weight loss
3. Insomnia, headach

Q-Drug interaction with theophylline

1. Drugs inhibit its metabolism toxicity a. Cimitidin b. Erythrobionin & quinololene c. Hypothyroidism , hepatic disease
2. Drugs accelerate its metabolism... decrease its efficacy.. a. Tomato, alcohol b. Antiepileptics :phenobarbital... c . Rifambicin d. Hyperthyroidism

Q-Adverse effects of theophylline

1. narrow safety margin
2. Irritant drug : gastritis if taken orally proctitis if rectally
3. Fast iv administration cause velocity reaction ... tachycardia, arrhythmia ,arrest, hypotension, syncope

- 1) How do β blockers improve cardiac function in HF ?
 - a- **By decreasing cardiac remodeling.**
 - a- By increasing heart rate.
 - b- By increasing renin release.
 - c- By activating norepinephrine.

- 2) K supplements should not be given if the following drug is prescribed:
 - a) **Enalapril**
 - b) Verapamil
 - c) Thiazide
 - d) Digoxin

- 3) First line of treatment of patients with elevated risk atherosclerotic CV diseases is:
 - a) **Statins**
 - b) Dabigatran
 - c) Alteplase
 - d) Ezetimibe

- 4) Early signs of digitalis toxicity is:
 - a) **Nausea & vomiting**
 - b) Yellow and green vision
 - c) Atrial arrhythmias
 - d) Heart block

- 5) Which is important to monitor in patients taking digoxin ?
 - a- Chloride.
 - b- **Potassium.**
 - c- Sodium.
 - d- Zinc.

- 6) Adverse effect associated with morphine includes:
 - a) Diarrhea
 - b) **Bronchoconstriction**
 - c) Lacrimation
 - d) Vasoconstriction

- 7) What is the most common adverse effect associated with fixed dose hydralazine/isosorbide dinitrate ?
 - a- Diarrhea.
 - b- Drug-induced lupus.
 - c- **Hypotension.**
 - d- Heartburn.

- 8) Variant angina is treated by the following combination:

A. Verapamil & Nitrate	C. Beta blocker & nifedipine
B. Beta blocker & nitrate	D. Nitrate & nifedipine

